# HCT in Elderly Patients

# 68

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## 68.1 Introduction

The hematological malignancies, which are the most common indications for auto- and allo-HCT (e.g., AML/MDS, NHL, MM, and others), are diagnosed at a median age greater than 65 years. Thus, if classical chronological age exclusion criteria were followed, a majority of patients with these malignancies would not be offered a HCT, despite it being their treatment of choice and in many cases their only curative option (Snowden et al. 2022). While elderly patients are more likely to face toxic effects from HCT, this risk must be considered and balanced against the poor outcome of transplant candidates with these malignancies who do not proceed to HCT (Dohner et al. 2022).

## 68.2 HCT Activity in Elderly Patients

Auto- and allo-HCT annual activity continues to steadily increase in Europe and worldwide with no signs of saturation (Gratwohl et al. 2015; Passweg et al. 2023). Specifically, in elderly patients, HCT activity at EBMT centers has increased markedly in the past two decades. Auto-HCT activity in patients ≥65 years old increased from 3.4% (443 out of 13,163 autologous HCT) in 2000 to 9.8% (2444 out of 23,883 auto-HCT) in 2014 (Sánchez-Ortega et al. 2016). Allo-HCT activity in patients ≥65 years old increased from <1% (37 out of 6413 allo-HCT) in 2000 to 6.7% (1057 out of 16,765 allogeneic HCT) in 2014 (Basak et al. 2016). Additional recent data demonstrated also the consistent increase in the use of auto-HCT in patients with MM, with the proportion of patients aged >65 years increasing dramatically, from 7% in the 90 s and first twenty-first century decade to 30% in the 2010s (Swan et al. 2022). This applies also for allo-HCT for NHL as per the EBMT registry, and the proportion of recipients aged 51–70 years increased from 8% in 1991–1995 to 58% in 2011–2015 (Kyriakou et al. 2019). For myelofibrosis, prior to 2006 only 8.7% of myelofibrosis patients undergoing allo-HCT were > 60 years, whereas it increased to 47% for recipients in the 2015–2018 cohort (McLornan et al. 2021).

In the USA, the number of auto-HCT and allo-HCT for the treatment of malignant diseases in patients aged ≥65 years continues to increase (D'Souza et al. 2020). In 2020, 27% of allo-HCT recipients were aged 65 years and older and 37% of auto-HCT recipients for lymphomas and MM

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were aged 65 years or older as compared to 4% for allo-HCT and 18% for auto-HCT in 2005. Moreover, trends in allo-HCT for AML in the last two decades show an increase in use in recipients aged 65 years and older (Auletta et al. 2021).

Improvements in supportive care, HSC mobilization, the use of RTC and RIC regimens, wider donor availability, including haploidentical for most patients, and the increase of newer therapeutic options improving remission status prior to HCT have contributed to the increase in HCT activity overall and, in particular, to the increase of HCT activity rates in elderly patients. With sustained improvement in these areas, and as the population ages, these numbers will only continue to increase.

## 68.3 HCT Outcomes in Elderly Patients

Compared to younger adults, elderly patients may have higher overall rates of transplant failure. Potential comorbidities, impaired health, and performance status could lead to higher transplant-related morbidity and mortality. In addition, malignancies in elderly patients often have more adverse disease features (e.g., higher-risk cytogenetics and molecular patterns in AML/MDS patients) and may have been treated less aggressively prior to HCT, which may potentially also increase the risk of disease relapse.

Historically, HCT outcome analysis in elderly patients has been limited by the fact that these patients are underrepresented in clinical trials and the majority of data come from relatively small series and subgroup analyses of small subsets of elderly patients in larger disease-specific studies including adults of all ages. More recently, HCT outcomes of elderly patients are being analyzed specifically and have reported feasibility and safety of autologous HCT in MM patients aged >65 years (Winn et al. 2015; Auner et al. 2015), in selected populations of elderly patients with R/R DLBCL (Chihara et al. 2014) and in R/R HL in patients aged ≥60 years (Martínez et al. 2017).

Prospective studies addressing the value of allogeneic HCT compared to non-transplant approaches are limited and generally restricted to patients aged <65 years. Interestingly, a recent study in a large series of AML patients aged 60 years and older suggests that age alone is not a barrier to successful HCT (Maakaron et al. 2022), and several large series in AML/MDS patients reported that NRM and OS were negatively affected by KPS <80–90% but not by chronological age (Heidenreich et al. 2017; McClune et al. 2010; Ringdén et al. 2019). Despite significantly poorer outcomes in older patients, additional trials have also not shown a significant impact of advanced age on major outcomes including NRM (Sorror et al. 2011; Chevallier et al. 2012).

The largest experience reported to date on auto- and allo-HCT outcomes in elderly patients comes from two EBMT studies including a total of 21,390 auto-HCT and 6046 allo-HCT in patients aged ≥65 years between 2000 and 2014 (Basak et al. 2016; Sánchez-Ortega et al. 2016). Patient numbers and key HCT outcomes overall and by age group are presented in Table 68.1.

These studies confirm the feasibility of auto- and allo-HCT in elderly patients, with acceptable NRM and OS at 1 and 3 years, respectively. Multivariate analyses in both studies showed that performance status (i.e., Karnofsky score) had a more significant independent impact on patient outcomes than chronological age. Thus, these data in a large cohort of elderly patients strongly suggest that age per se should not be an exclusion criterion to consider HCT in this population. Undoubtedly, this is presumably a highly selected

**Table 68.1** HCT outcomes in elderly patients: EBMT experience[a]

| Type of HCT | All cases ≥65 years | Group I 65–69 years | Group II ≥70 years |
|---|---|---|---|
| Autologous, *n* | 21,390 | 17,531 | 3859 |
| – NRM year 1 | 4.9% | 4.6% | 5.9% |
| – OS year 1 | 87% | 88% | 83% |
| – OS year 3 | 67% | 69% | 61% |
| Allogeneic, *n* | 6046 | 4914 | 1132 |
| – NRM year 1 | 27% | 26% | 29% |
| – OS year 1 | 57% | 57% | 53% |
| – OS year 3 | 39% | 40% | 35% |

*n*, number of cases.
[a]Basak et al. (2016) and Sánchez-Ortega et al. (2016)

fraction of elderly patients considered for auto- and allo-HCT. Nevertheless, this further endorses the need to assess comorbidity and frailty beyond the age in older HCT candidates to improve outcomes further.

## 68.4 Assessment of Elderly Candidates for HCT

In addition to the elements already discussed in Chap. 11 for younger patients, the evaluation and counseling of elderly patients as candidates for auto- and allo-HCT require the evaluation of additional health domains of interest in patients of advanced age. The following tables address general principles and considerations for the evaluation and counseling of these patients, discuss the issue of patient frailty beyond age and comorbidities, and describe the key elements of a multidimensional geriatric assessment in this population.

### 68.4.1 General Principles and Considerations for Elderly HCT Candidates

- HCT decision *should not be driven by chronological age* but by a broader multidimensional assessment, including fitness, comorbidities, physiologic reserve, and frailty.
- Elderly patients require *information and counseling in plain language* regarding the HCT process, donor sources, specific protocol, timeline, risks, benefits, and outcomes.
- They also need information regarding patients' *quality of life outcomes, caregivers, and psychosocial needs*, for which social workers and other support staff will be needed.
- A *multidisciplinary individualized assessment* is required to appropriately address the multidimensional nature of the evaluation of elderly patients.
- Fit older transplant candidates should follow the *same indications for auto- and allo-HCT as younger adults*.
- In the case of allo-HCT, particular consideration of *RIC and NMA regimens* is essential, and donor selection must take into account the *age of the donor*, as donor older age may associate with impaired outcomes.
- Outcome analysis in elderly patients may require the use of *clinically relevant composite endpoints* that, beyond survival, incorporate quality of life, good overall mental and physical condition, and freedom from severe complications.

### 68.4.2 Frailty in Elderly HCT Candidates

- Frailty is a term used to describe a multidimensional syndrome of loss of physiologic reserves (energy, physical ability, cognition, and health) that leads to vulnerability.
- The ability to measure frailty in elderly patients is useful clinically.
- Although it appears to be a valid construct to assess elderly patients, how exactly to define it remains unclear. There is a large abundance of possible scales to measure frailty, which likely reflects uncertainty about the term and its components.
- A. Hedge et al. have recently reported on frailty as the missing piece of pre-HCT assessment (Hegde and Murthy 2018). Data show that the prevalence of frailty prior to HCT in patients aged ≥50 years is higher than in the general geriatric population at around 25%. Importantly, age has no effect on the prevalence of frailty.
- Frailty is associated with poorer OS even after adjusting for age and HCT-CI and may be associated as well with an increased incidence of disease relapse (Muffly et al. 2014; Hegde and Murthy 2018).

### 68.4.3 Geriatric Assessment for Elderly HCT Candidates

#### 68.4.3.1 General Concept

- The geriatric assessment is a multidimensional, multidisciplinary assessment designed to evaluate an older person's functional ability, physical health, cognition, mental health, and socio-environmental circumstances (Artz 2016).
- The goal of geriatric assessment in this context would be to capture vulnerability pre-HCT to help deciding on patient suitability for the procedure as well as to individualize post-HCT support strategies to prevent complications and reduce transplant-associated morbidity and mortality (Artz 2016; Jayani et al. 2020).

#### 68.4.3.2 Elements Involved in Elderly HCT Candidates

- Ensure appropriate performance status (Karnofsky score $\geq 80$).
- Rule out significant comorbidities by the HCT-CI (Sorror et al. 2005), as their prevalence increases with age.
- Assess the modified EBMT (Armand et al. 2014) and the revised PAM scores (Au et al. 2015), as global prognostic models that incorporate both NRM and disease factors.
- Measure functional status by self-reported function and performance-based testing (ability to perform tasks necessary to live independently in the community [i.e., shopping, food preparation, housekeeping, telephone, laundry, transportation and driving, manage finances and medication, number of times a patient can rise from the chair (i.e., timed up and go), gait speed, 6-min walk test, hand grip strength, or provocative cardiopulmonary testing], polypharmacy requirements).
- Cognitive function: if necessary, perform neuropsychological testing and/or consult geriatrics (Oli et al. 2020).
- Psychosocial evaluation (assessment of social support, availability of a caregiver, financial matters, psychological disturbances, etc.).
- Nutritional status and weight loss.
- Biomarkers to characterize physiologic age (serum C-reactive protein, ferritin, serum albumin, or protein biomarkers panels in development).

#### 68.4.3.3 Scales

- No standard geriatric assessment scales have been validated for HCT.
- Most scales available for geriatric assessment in cancer patients are complex and time-consuming, which limits its use in daily practice.
- The Geriatric Assessment in Hematology (GAH) scale is a brief, comprehensive geriatric assessment scale designed and validated for older patients diagnosed with hematological malignancies (MDS, AML, MM, and CLL) (Bonanad et al. 2015).
- The GAH scale has been shown to be responsive to clinical changes in patient's health status (Cruz-Jentoft et al. 2017) and may also help discriminate patients who will benefit most from intensive treatments from those requiring an adapted approach (de la Rubia et al. 2023).
- The GAH scale includes 30 items grouped into 8 predefined dimensions (number of drugs, gait speed, mood, activities of daily living, subjective health status, nutrition, mental status, and comorbidities) and requires a relatively short period of time to be administered in routine clinical practice (10–12 min).
- Thus, the GAH scale could be an interesting tool to assess elderly patients with hematological malignancies who are being considered for transplantation. However, it still needs to be validated in the setting of HCT.

**Key Points**

- HCT activity in elderly patients has increased markedly in the past two decades and is predicted to continue to increase as the population ages, with a sustained improvement in HCT methodology and care.

- Auto- and allo-HCT in elderly patients is feasible and has acceptable outcomes.
- Age should not be an exclusion criterion per se to consider elderly patients for HCT.
- Assessing comorbidity is essential in older HCT candidates, but adjusting only for comorbidity may not identify frail patients vulnerable to adverse outcomes.
- Frailty is a multidimensional syndrome of loss of physiologic reserves (energy, physical ability, cognition, and health) that leads to vulnerability, is higher in HCT recipients than the general geriatric population, and associates with poorer HCT outcome.
- Geriatric assessment is a multidimensional, multidisciplinary assessment designed to evaluate an older person's functional ability, physical health, cognition, mental health, and socio-environmental circumstances.
- The goal of geriatric assessment in HCT would be to capture vulnerability pre-HCT to help deciding on patient suitability for the procedure and to adapt post-HCT support strategies to improve outcomes.
- Currently, there are no standard geriatric assessment scales validated for HCT. The GAH scale has been described and validated in elderly patients with hematological malignancies, is relatively simple to apply in clinical practice, and may be a candidate scale for elderly HCT candidates, validation pending.

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